Lab Experiment-20

Aim

To write a Python program to implement a basic Information Retrieval System using TF-IDF (Term Frequency-Inverse Document Frequency) to rank documents based on their relevance to a user query.

Algorithm

1. Import the required libraries.
2. Create a collection of sample documents.
3. Get a search query from the user.
4. Convert the documents into TF-IDF vectors using `TfidfVectorizer`.
5. Convert the user query into a TF-IDF vector.
6. Calculate the similarity between the query and documents using cosine similarity.
7. Rank the documents based on their similarity scores.
8. Display the ranked documents.

Python Code

```
from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.metrics.pairwise import cosine_similarity

documents = [

    "Python is a programming language",

    "Machine learning is a part of artificial intelligence",

    "Python is used for machine learning",

    "Artificial intelligence is the future"
```

```
]

query = input("Enter your search query: ")

vectorizer = TfidfVectorizer()

tfidf_matrix = vectorizer.fit_transform(documents)

query_vector = vectorizer.transform([query])

scores = cosine_similarity(query_vector, tfidf_matrix)

ranking = scores[0].argsort()[::-1]

print("\nRanked Documents:")

for i in ranking:

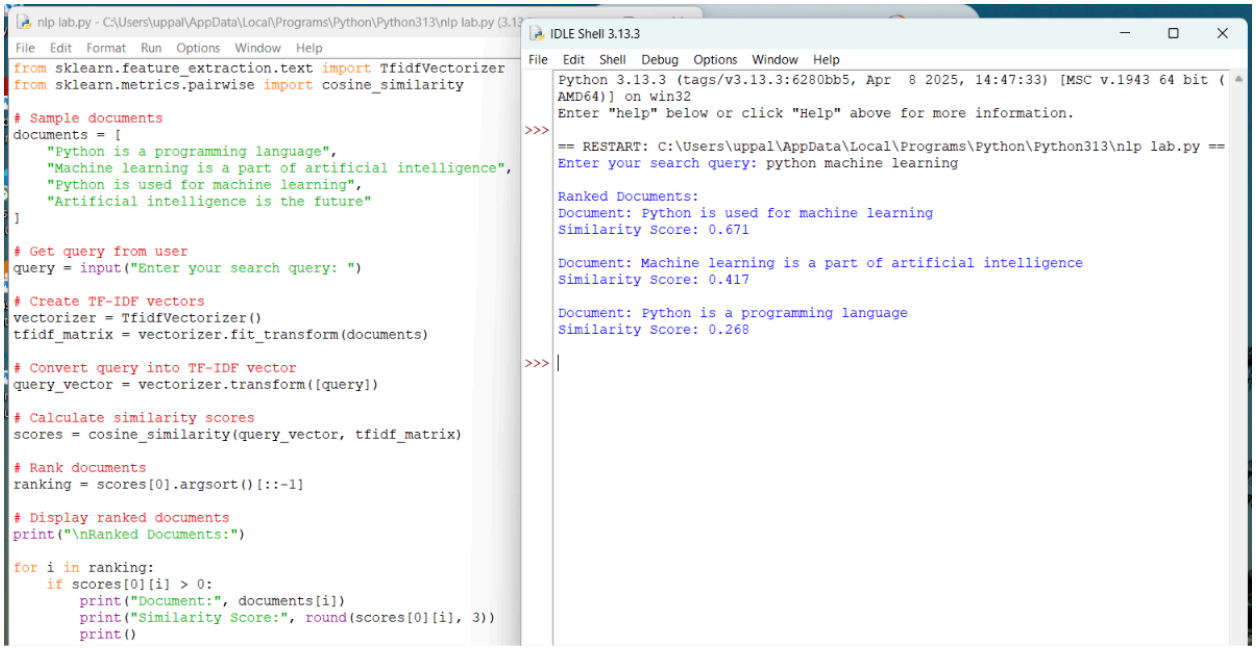
    if scores[0][i] > 0:

        print("Document:", documents[i])

        print("Similarity Score:", round(scores[0][i], 3))

        print()
```

Output:



Result

Thus, the Python program successfully implements an Information Retrieval System using TF-IDF and cosine similarity to rank documents according to their relevance to the user's search query.